

Identifying Trends: The Impact of Economic and Meteorological Factors on Suicides in Chile for All Age Groups

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Abstract

This paper explores the complex connections between weather, economic indicators, and suicide rates across four different age groups. Utilizing the random forest method for imputation of missing values and applying Poisson regression models, the study examines the impact of temperature, atmospheric pressure, precipitation, humidity, unemployment rate, and inflation rate on suicide rates. The analysis is much more robust because of the use of multiple imputations, and it is found that whereas economic variables show significant connections, climatic elements are not important predictors. The study emphasizes the sensitivities among different age groups and highlights the significance of focused strategies and accurate evaluations of risk to prevent suicide.

Note: All the relevant code used for this study is given in the Appendix section of the paper.

Keywords

Meteorological Factors, Economic Indicators, Multiple Imputation, Poisson Regression, Suicide Rates, Age Groups

Introduction

The current study focuses on the analysis of completed suicides that occurred in Chile in 2020. It attempts to further examine the potential influence of economic indicators

and bioclimatic factors, as referenced in existing literature (Brenner, de Almeida Claudino, de Souza, & da Rocha., 2022), on the occurrence of suicides across different age groups. The importance of this investigation is highlighted by concerning statistics, which reveal that almost 800,000 people die by suicide each year, and more than 20 suicide attempts occur for each death (Kim et al., 2019), thus categorizing suicide as a pressing global public health concern.

A comprehensive review of the literature highlights varied perspectives on the impact of bioclimatic factors on completed suicides. Early studies suggested a challenge in establishing a clear correlation between specific weather conditions and suicide rates, potentially due to the high variance in the methodological approaches of the different studies (Deisenhammer, 2003). However, a subsequent meta-analysis contradicted this finding, identifying a positive correlation between temperature and completed suicides (Gao et al., 2019). Recent research further emphasizes the importance of bioclimatic parameters, especially when it comes to the population of people 60 years of age and older (Kim et al., 2019).

In terms of economic indicators, conflicting findings contribute to the uncertain relationship with completed suicides. An older study identified a connection between unemployment and suicide attempts, but not between completed suicides and unemployment (Solano et al., 2011). Conversely, a different study reported a positive association between inflation and completed suicide (Lari, 2023).

Transitioning from the literature review to the dataset selection, the decision to focus on Chile offers distinct advantages. Specifically, choosing a data set from one country lightens our analysis of the challenges posed by missing values in large data sets. This focused approach empowers us to concentrate entirely on our research questions. Notably, the selected country is the longest in the world from north to south and that offers a remarkable opportunity to delve deeper into the association between completed suicides and economic indicators, as well as completed suicides and meteorological factors taking also into account the geographical location.

Overall, this study focuses on two major research questions. Firstly, it aims to investigate the impact of economic factors on the number of completed suicides. Secondly, this analysis intends to study the effect of meteorological conditions on the number of completed suicides. It will additionally be examined for both questions, whether the specific associations differ across different age groups, and only for the second question whether it differs across geographical locations in Chile in 2020.

Methods

A methodical research approach was used to unravel the intricate interactions among completed suicides and economic indicators as well as bioclimatic conditions in Chile in 2020. This section describes the methodology used in the selection of participants, the data gathering and description, and the approach applied to each study question.

Participants and design

The sampling method, which employs a census approach, is designed with precision to encompass a nationwide perspective, including all recorded cases of completed suicides throughout the given year in Chile. Our primary data set consists of 1602 observations (suicides) and is provided by National Statistics Institute (Chile)(*National Statistics Institute (Chile)*, n.d.). Since, as already mentioned, Chile is a very long country, it is divided into Regions, Provinces, and Municipalities, which are all provided in the dataset, for each suicide. Each Region includes many Provinces and each Province includes many Municipalities. This information was crucial for understanding the measurements in the primary dataset. So, the dataset has been enriched with detailed information on daily meteorological factors different for every Municipality and monthly economic indicators, same across every Municipality.

The meteorological data were collected from local meteorological stations near the suicide locations, providing average values, for the day of each suicide, within the specific Municipality. More specifically, we used weather data from the Dirección Meteorológica de Chile (*Dirección Meteorológica de Chile*, n.d.). Additionally, monthly economic indicators were obtained from Central Bank of Chile (*Central Bank of Chile*, n.d.), featuring one value per month, consistent across all Municipalities of Chile. To enable an in-depth analysis, the sample is split into four age groups. More specifically, these age groups include individuals under the age of 26 (305 observations), those between the ages of 26 and 40 (455 observations), those between the ages of 40 and 60 (512 observations), and those over the age of 60 (332 observations). The terms "Adolescents", "Young Adults", "Middle-aged Adults", and "Elderly Adults" are used to denote the four aforementioned age groups, respectively. This stratification allows for a more detailed examination of completed suicides across different life stages, leading to a broader comprehension of the phenomenon.

Proceeding further to the participants in this study are individuals who died by suicide in Chile in 2020. Individuals of all ages who committed suicide within the specified

year are included in the criteria, which take into consideration all available cases from municipalities across the country. Hence, this study employs a retrospective observational design to investigate the relationship between economic indicators, bioclimatic factors, and completed suicides across various age groups and geographical locations. The study is cross-sectional and exploratory, as we examine data for a single point in time, the time of the suicide, and we attempt to understand the relationships between economic and bioclimatic factors with completed suicides without making causal relationship assumptions. To succeed in this, the variables under study include monthly unemployment and inflation rates for economic indicators, and daily mean temperature, humidity, atmospheric pressure, and precipitation for bioclimatic factors, in the way that was previously described.

Measurements

Moving on, a thorough synopsis of all the factors used in the research is provided. The intention is to clarify the definitions of these variables and describe the methods used for working with them. The variables include a variety of parameters linked to suicide incidents, beginning with "Age," which denotes the age of the participants at the time of suicide, expressed in years. "Day of Death" specifies the numerical day value of the month when the suicide occurred. Similarly, "Month of Death" represents the numerical value of the month, while "Year of Death" is recorded as a four-digit numerical value. To examine the variations, regarding the research questions, across different geographical locations, only the variable "Region" was considered, featuring 16 regions in the dataset. Consequently, columns containing information on Municipalities and Provinces were removed from the initial dataset, as they were deemed irrelevant to the analysis. The motivation behind this action was to simplify the dataset, enhancing overall clarity and interpretability in the subsequent analysis.

To continue describing the data used for the study, meteorological variables were assessed daily, as already mentioned. These include "Mean Temperature," "Mean Atmospheric Pressure," "Precipitations," and "Humidity". These variables represent the average temperature (in degrees Celsius), atmospheric pressure (in hectopascals), precipitations (in millimeters), and humidity (in %) for the specific day and municipality. Additionally, economic factors were measured monthly for each municipality and encompass the "Unemployment Rate" and "Inflation Rate," both expressed as percentages (%). These measures show, correspondingly, the proportion of the local population without a job and the percentage change in the average level of prices for goods and

services.

The aforementioned variables were utilized to examine their relationship with the number of suicides. The techniques used to do that will be explained in the section that follows. Before delving into that though, it is crucial to note that the data set, encompassing those variables, underwent pre-processing. Two new data sets, employed for the subsequent analysis, were generated as a result. Both data sets incorporate a column for "Age Group". However, the first dataset comprises daily entries, while the second encompasses monthly entries for the year 2022. The initial dataset contains average meteorological measurements for days when suicides occurred throughout the year. It includes separate entries for each region where these incidents took place on specific days. Each entry for a particular region corresponds to a distinct age group associated with a suicide occurrence, for which the mean across the suicides within the specific age group is calculated. It is important to note that for the same Region and same day, different mean values might occur, for each age group, because of the fact that a region consists of many municipalities each one of them having different average daily measurements. The second dataset is simpler since for all the regions, the same economic indicators are provided per month. So, it contains the average economic variables across all regions for every month. Each one of the datasets was used for the respective research query.

Statistical analysis

Regarding the statistical analysis employed in this study, as already mentioned, it aims to investigate the relationship between meteorological and economic factors and the number of completed suicides across different age groups. To address the research questions, **Poisson regression** analysis was chosen, keeping in mind the count nature of the outcome variable (number of suicides) and the relevance of predictor variables. Two separate analyses were conducted, using the daily meteorological data set and the monthly economic data set, with the already mentioned design.

For the analysis concerning meteorological factors, it was considered important to account for the variations across the different geographical locations. To do so, standardizing measurements within each Region were used, by utilizing z-scores for all the meteorological factors. The z-score not only standardized the variables but also preserved necessary locale-based information about the values of these variables. Hence, we mitigated the risk of losing valuable insights associated with different regions while

taking into account their in some cases different profiles. Hence, two Poisson regression models were fitted with the number of suicides as the dependent variable in both of them. In the first one, the z-scores of temperature, atmospheric pressure, humidity, and precipitation were the predictor variables and in the second one, the interactions of them with age were additionally considered.

On the same pattern, for the analysis in relation to economic factors, two Poisson regression models were fitted with the number of suicides again as the response variable in both of them and the unemployment and inflation rate as the predictors in the first one, but in the second one with the interactions of age with them as well.

When employing Poisson regression analysis, there are several key assumptions that need to be carefully considered. These assumptions determine the validity and reliability of the study's findings. First, it was assumed that suicides have typically happened on their own, independently, and at a steady average rate. Homogeneity of variance among the various levels of the independent variables was also presumed. Additionally, over-dispersion was considered absent since that ensures the alignment of variance and mean of the dependent variable. Small differences though, could be neglected for relatively big sample sizes as in our study.

Therefore, for diagnostic purposes, two chi-square tests were employed to evaluate the overall performance of the fitted model: the Pearson statistic and the deviance statistic. Both of these statistics follow an approximate chi-square distribution with degrees of freedom equal $n-k-1$, where n is the sample size, and k is the number of estimated parameters. It is worth highlighting that, the Pearson statistic, used to measure goodness-of-fit, serves a dual purpose by also acting as a test for overdispersion. This is especially relevant in scenarios where the observed data's variance exceeds that predicted by the given model for $Y | X$ (Y : number of suicides, X : predictor variables). Thus, a significant rejection of these tests (p -values < 0.05) suggests a statistically meaningful lack of fit in the model, while non-significant results (p -values > 0.05) indicate the absence of evidence for a lack of fit (Eberly, n.d.).

Missing values were present in the original dataset, used for the analysis, so this problem needed to be carefully addressed before continuing with the previously described analysis. Following our initial data inspection, we continued to form assumptions about the missing data. It is noteworthy that the only variable that raises concerns due to a significant percentage of missing data is the precipitation variable. For the rest of the variables, the percentage of missingness is lower than 2%. Consequently, we assumed that the missing values were Missing Completely At Random (MCAR), indicating that

the absence of data had no association with any identifiable participant characteristics or other variables within our data set.

However, specifically for the precipitation variable, we confirmed the absence of the precipitation variable by contacting the relevant meteorological stations. Our research revealed that the lack of data was due to technical errors in which measurements were not recorded because of a machine breakdown. Hence, to be able to include the precipitation variable in our analysis, we decided to use the multiple imputation method. To be able to use this method, we assumed that the existing variables in our data set could be used as relevant factors to predict missing values as accurately as possible. This assumption holds as we have variables such as the date, the region, the temperature, the atmospheric pressure, and the humidity that could be shared information between the completed rows and those with the missing value for precipitation. The imputation model we developed produces five completed data sets (having no missing values) in a manner consistent with the observed patterns in our real data set and imputed all the missing values, even for those with lower than 2% missingness. Aiming for robust predictions, we employed the random forest method in the imputation process, which effectively included key predictor variables that will predict the missing values as precisely as possible. Furthermore, to improve imputation accuracy, variables with low correlations (minor 0.1) were excluded and certain contextual variables were taken into account to capture complicated relationships. Subsequently, we aggregated the results of individual analyses on each imputed data set into a pooled outcome which is the closest to what the result would have been if we had no significant percentage of missing data. Overall, the multiple imputation method significantly improves the robustness of our analysis and is necessary for an accurate examination of the precipitation variable.

Software The R statistical programming language, along with relevant packages for data manipulation, regression modeling, and diagnostic evaluations, was used to perform the statistical studies. R was chosen since it offers a stable and adaptable framework for statistical analysis that guarantees the results' reproducibility and transparency.

Results

Multiple Imputation

Before delving into the Poisson regression analysis findings, we first explored the descriptive statistics for the main variables in our research questions. Table 1 provides information on the mean and standard deviation for the six variables, before and after multiple imputation. Consequently, the multiple imputation had a minor effect on both the central tendency and variability of the variables related to the meteorological effects and no effect on the economic indicators as we expected based on the fact we had no missing values in them.

Variable	Mean (Before)	Std (Before)	Mean (After)	Std (After)
Temperature	14.709	4.957	14.741	4.956
Atmospheric Pressure	975.002	70.539	975.129	69.926
Precipitations	1.162	4.897	0.983	4.428
Humidity	69.903	16.393	69.824	16.401
Unemployment rate	0.104	0.020	0.104	0.020
Inflation rate	0.003	0.003	0.003	0.003

Table 1: Descriptive Statistics for the Variables in Interest, Before and After Multiple Imputation

Regression Analysis

Since our main analysis included two main research questions, we will divide this section accordingly and approach each of our queries separately.

Meteorological Factors

To begin with, a series of Poisson regression models were fitted to each imputed data set to examine the relationship between meteorological factors and the number of daily suicides while accounting for regional differences and the across age groups difference. More specifically, the temperature, atmospheric pressure, precipitation, and humidity were initially studied, and further by adding their interactions with the four age groups.

As we highlighted previously, taking into account the potential regional variations, we incorporated random intercepts for regions.

As exhibited in Table 2, the analysis of meteorological factors captured by the generalized linear mixed model reveals interesting insights into their associations with daily suicide rates. The intercept, representing the baseline, has a statistically significant positive association with suicide rates (Estimate = 0.0725, p-value = 0.0135* at level $\alpha=0.05$), indicating only that the log of the mean number of suicides (at the value 0 of the other covariates) is different from 0. Thus, none of the meteorological variables exhibits a significant effect on the suicide counts per day.

Variable	Estimate	Std. Error	z value	p-value
(Intercept)	0.0725	0.0203	2.469	0.0135*
Temperature	0.0174	0.0319	0.544	0.5864
Atmospheric pressure	0.00006	0.02661	0.002	0.9983
Precipitations	-0.000009	0.02643	0.000	0.9997
Humidity percentage	0.0007	0.03292	0.021	0.9829

Table 2: Regression Results for Meteorological Factors model without taking age group into account

When each meteorological variable is examined closely with their interaction terms (Table 3), the temperature has no significant association with suicide rates across all age groups. Moreover, variations in atmospheric pressure, precipitation, and humidity have similarly non-significant relationships with suicide rates, implying that these variables may not be strong predictors of daily suicide occurrences. Once again, we observe a significant connection with the baseline conditions (Intercept, Estimate = 0.0741, $p = 0.0134^*$ at level $\alpha=0.05$) which means that the suicide rates are not efficiently explained across age groups by any of the meteorological factors we included in our regression model. Overall, there was no statistically significant interaction between meteorological factors and age groups, indicating that the impact of weather on suicide rates is consistent across age groups.

Economic Indicators

Moving on to our second research question, we examined how economic indicators, specifically the unemployment rate and the inflation rate, affected monthly suicide rates. In line with our previous analysis, Poisson regression models were utilized in

Variable	Estimate	Std. Error	z value	p-value
(Intercept = baseline) 0.0729	0.0294	2.484	0.013*	
Temperature:Adolescents	0.0322	0.0721	0.447	0.655
Temperature:Young Adults	0.0272	0.0570	0.477	0.633
Temperature:Middle-age Adults	0.0109	0.0628	0.174	0.862
Temperature:Elderly Adults	0.0006	0.0673	0.009	0.993
Atmospheric pressure:Adolescents	0.0132	0.0607	0.217	0.828
Atmospheric pressure:Young adults	-0.0266	0.0568	-0.467	0.640
Atmospheric pressure:Middle-age Adults	0.0188	0.0480	0.391	0.696
Atmospheric pressure:Elderly Adults	-0.0077	0.0509	-0.151	0.880
Precipitations:Adolescents	0.0104	0.0885	0.117	0.907
Precipitations:Young Adults	-0.0021	0.0426	-0.049	0.961
Precipitations:Middle-age Adults	-0.0028	0.0490	-0.058	0.954
Precipitations:Elderly Adults	-0.0038	0.0568	-0.068	0.946
Humidity (:Adolescents)	-0.0047	0.0724	-0.065	0.948
Humidity:Young Adults	0.0151	0.0588	0.257	0.797
Humidity:Middle-age Adults	-0.0036	0.0651	-0.055	0.956
Humidity:Elderly Adults	-0.0059	0.0713	-0.083	0.934

Table 3: Regression Results for Meteorological Factors Across Age Groups

this analysis to capture the count nature of the suicide data. Additionally, to investigate potential differences across age groups, we included interaction terms with each of these economic indicators in interest.

In Table 4 we display the findings of Poisson regression analyses that evaluated the relationship between these factors and the number of suicides per month without taking into consideration the age group interaction. Observing Table 4, we can conclude that the intercept has a statistically significant positive relationship with monthly suicide rates (Estimate = 3.8970, p-value < 0.001*** at level $\alpha=0.05$). The interpretation of this significance is that a baseline level of risk remains influential, regardless of the economic factors included in the model. Proceeding further to the evaluation of the significance of the variables in interest, we have a significant effect of both the unemployment rate and inflation rate on the monthly counts of suicides. In particular, a higher unemployment rate percentage is related to fewer daily suicides (Estimate = -4.4748, p-value = 0.000607 *** at level $\alpha=0.05$), whereas a higher inflation rate percentage is related to higher numbers of daily suicides (Estimate = 31.7419, p-value = 0.000583 *** at level $\alpha=0.05$).

Variable	Estimate	Std. Error	z value	p-value
(Intercept)	3.8970	0.1459	26.711	< 2e-16 ***
Unemployment Rate (%)	-4.4748	1.3052	-3.428	0.000607 ***
Inflation Rate (%)	31.7419	9.2294	3.439	0.000583 ***

Table 4: Poisson Regression Results for Economic Indicators

Finally, Table 5 displays the Poisson regression results for the impact of economic indicators on monthly suicide rates in various age groups. The intercept has a significant positive association with suicide rates once again, whereas the inflation rate has no apparent impact across all age groups. The unemployment rate, on the other hand, seems to have distinct associations across age groups. Specifically, higher unemployment rates among young adults and middle-aged adults are linked to higher suicide rates, highlighting the importance of economic factors in these age groups. In particular, the estimate for the higher unemployment rate among young adults is 3.3822 (p-value = 0.000165 *** at level $\alpha=0.05$), while the estimate for middle-aged adults is 3.8968 (p-value = 1.12e-05 *** at level $\alpha=0.05$). Surprisingly, a significant negative association is found for adolescents (Estimate = -6.7665, p-value = 3.18e-06 *** at level $\alpha=0.05$), implying that higher unemployment rates are associated with lower suicide

rates in this age group.

Variable	Estimate	Std. Error	z value	p-value
(Intercept)	3.9188	0.1460	26.843	< 2e-16 ***
Inflation Rate:Adolescents	13.2521	19.6602	0.674	0.5003
Inflation Rate:Young Adults	14.2100	24.8707	0.571	0.5678
Inflation Rate:Middle-age Adults	33.8502	24.3143	1.392	0.1639
Inflation Rate:Elderly Adults	17.3653	26.5422	0.654	0.5130
Unemployment Rate:Adolescents	-6.7665	1.4525	-4.659	3.18e-06 ***
Unemployment Rate:Young Adults	3.3822	0.8976	3.768	0.000165 ***
Unemployment Rate:Middle-age Adults	3.8968	0.8872	4.392	1.12e-05 ***
Unemployment Rate:Elderly Adults	0.4332	0.9679	0.448	0.6545

Table 5: Poisson Regression Results for Economic Indicators Across Age Groups

Diagnostics

Performing the previously mentioned Diagnostic Procedures of the two chi-square tests (Pearson and Deviance statistics) the results proved encouraging for the majority of our analyses. To delve deeper, let's explore the performance of the four models discussed in this paper.

Models for Meteorological Factors

The Deviance and Pearson Chi-Square values for the model regarding meteorological factors, without taking into account the Age group variable, were calculated to be 91.81694 and 119.2369, respectively, with associated p-values of 1.

For the model in which the Age group was considered, the Deviance Chi-Square value was calculated to be 91.41812, whereas the Pearson Chi-Square value was calculated to be 118.451. Both were again accompanied by p-values of 1.

These results suggest a significant lack of evidence for lack-of-fit as well as for overdispersion, for both the models. These results, demonstrate that our models effectively capture the observed data and account for the variability present in the dataset.

Models for Economic Indicators

In relation to the model for the economic indicators, without taking into account the Age Group, the results were not that positive. Specifically, the Deviance and Pearson Chi-Square values were 108.4543 and 111.3796, correspondingly, both associated with extremely low p-values (3.71×10^{-7} and 1.5×10^{-7}). These findings point to a statistically significant lack of fit in the model, indicating that the observed data patterns are not well captured by the model. Furthermore, the existence of statistically significant overdispersion suggests that the data's variability is higher than predicted by the assumed model.

On the other hand, the results for the model in which Age Group was taken into account (with 39 degrees of freedom for residuals) the results of the Goodness-of-Fit Tests, utilizing Deviance and Pearson statistics, indicate a lack of statistically significant evidence for lack-of-fit and overdispersion. The Deviance and Pearson Chi-Square values were 40.23106 and 40.74102, respectively, with associated p-values of 0.4155348 and 0.3937445. These non-significant p-values imply that there are no significant differences between the expected model and the observed data, showing that the observed data are in good agreement with the model. Moreover, the absence of evidence of overdispersion implies that the variability in the data is consistent with what is expected under the assumed model.

Discussion

Summary

In summary, the Poisson regression analyses were conducted to investigate the relationship between meteorological and economic factors and the number of completed suicides separately and additionally taking into consideration four different age groups. For meteorological factors, including temperature, atmospheric pressure, precipitation, and humidity, the results showed no significant associations with the number of suicides per day. The inclusion of interaction terms with the age groups also indicated non-significant results, indicating that the weather has an even impact on suicide rates for all age groups. On the other hand, economic indicators such as the unemployment rate and inflation rate yielded significant effects on the suicide rates per month. Interestingly, the impact of the unemployment rate varied across age groups, with significant associations found for adolescents, young, and middle-aged adults.

To relate the research questions with the results, they are focused on figuring out how meteorological and economic aspects are related to suicide rates and how these differ across different age groups. So, in this study, meteorological variables did not exhibit statistically significant correlations with suicide rates. Contrary, economic indicators, and especially the unemployment rate for younger people, were highly significant. These findings highlight the importance of considering age-specific vulnerabilities for a better understanding of the complex interactions between many variables and the number of suicides.

Implications

The implications of the study's findings vary for public health and guidelines that should be considered to reduce suicide rates. Understanding the impact of economic indicators on suicides emphasizes the need for specific initiatives, particularly when economic instability is high. Moreover, the absence of significant associations with meteorological factors implies that the influence of other socioeconomic and mental health factors might be bigger for suicide rates.

Connection with other studies

The patterns our analysis found are consistent with some of the literature that has already been written, as this paper's introduction indicated. Previous studies have shown economic parameters that affect suicide rates, highlighting the reliability of these findings (Lari, 2023). Some studies also find non-significant relationships with meteorological parameters, which highlights the complex relationship of factors contributing to suicidal behaviors (Deisenhammer, 2003).

Limitations

Understanding the limitations of the study is necessary for a fair interpretation of the results. Firstly, even though the multiple imputation method shows robustness, it also might introduce bias because it relies on several assumptions that may not hold. Furthermore, the study's concentrated attention on certain variables might have ignored other important economic and meteorological parameters on suicide rates, such as GDP for economic factors or sunlight for meteorological aspects. Finally, the poor diagnos-

tics obtained when the Poisson regression model was applied to economic conditions without taking the age group into account, indicate inaccuracies in its results.

Future work

In light of the preceding limitations, future research projects may explore more socio-economic and meteorological variables in addition to a thorough investigation of mental health concerns. Hopefully, with this approach, a more comprehensive knowledge of the factors that contribute to suicide will be gained. In addition, we suggest that the acquired data for future research should be for all the days of the considered period, rather than concentrating just on the days on which suicides happened. Making this modification will allow for meaningful comparisons. Notably, although the country of choice may provide insightful information, because of its length and variety of meteorological conditions, collecting data over a longer time frame is advised to guarantee there is enough data for regional studies.

Conclusion

In conclusion, this study sheds light on the intricate relationships between economic indicators, meteorological factors, and suicide rates across different age groups. While economic indicators demonstrated significant associations, meteorological factors were not proven as important predictors by this study. Despite their limitations, our results add to the existing discussion on suicide prevention by highlighting the importance of targeted strategies and thorough risk evaluation techniques. To inform more effective public health projects, future research projects should delve deeper into the complex interactions of factors impacting suicide rates.

Appendix

Code for the research as Rmd file

Research Questions: Firstly, we aim to investigate the impact of economic factors on number completed suicides. Secondly, the study intends to study the effect of meteorological conditions on the number of completed suicides. Both questions will be examined across the different age groups in Chile in the year 2020. Additionally, the geographical location is also taken into account for the research question regarding meteorological factors.

Import Libraries

Read Data

```
original_data <- read_excel("Dataset_CS_2020_copia.xlsx", col_types = "text")
print(head(original_data))

## # A tibble: 6 x 21
##   Seks Age 'Day of death' 'Month of death' 'Year of death'
##   <chr> <chr> <chr>          <chr>          <chr>
## 1 1    20    1              1              2020
## 2 1    62    1              1              2020
## 3 2    63    1              1              2020
## 4 1    57    1              1              2020
## 5 1    53    1              1              2020
## 6 1    51    1              1              2020
## # i 16 more variables: 'Municipality code' <chr>, Region <chr>, Province <chr>,
## #   Municipality <chr>, 'Diagnostic 1 (cause of death)' <chr>,
## #   'Diagnostic 2 code' <chr>, 'Diagnostic 2 (suicide method)' <chr>,
## #   'Place of occurrence' <chr>, 'Lowest Temperature' <chr>,
## #   'Highest Temperature' <chr>, 'Mean temperature' <chr>,
## #   'Athmosferic pressure (mean) in hPa' <chr>, 'Precipitations (mm)' <chr>,
## #   'Humidity (%)' <chr>, 'Unemployment rate (%)' <chr>, ...

nrow(original_data)

## [1] 1602
```

Preprocess Data

```
#change column names
colnames(original_data) <- str_replace_all(colnames(original_data), " ", "_")

colnames(original_data)
```



```
## [1] "Seks" "Age"
## [3] "Day_of_death" "Month_of_death"
## [5] "Year_of_death" "Municipality_code"
## [7] "Region" "Province"
## [9] "Municipality" "Diagnostic_1_(cause_of_death)"
## [11] "Diagnostic_2_code" "Diagnostic_2_(suicide_method)"
## [13] "Place_of_occurrence" "Lowest_Temperature"
## [15] "Highest_Temperature" "Mean_temperature"
## [17] "Athmosferic_pressure_(mean)_in_hPa" "Precipitations_(mm)"
## [19] "Humidity_(%)" "Unemployment_rate_(%)"
## [21] "Inflation_rate_(%)"
```

```
#Remove columns we don't need
```

```
columns_to_remove <- c("Year_of_death", "Municipality", "Diagnostic_2_code",
  "Place_of_occurrence", "Highest_Temperature", "Seks",
  "Municipality_code", "Province", "Diagnostic_1_(cause_of_death)",
  "Diagnostic_2_(suicide_method)", "Lowest_Temperature")
```

```
data <- original_data[, !(names(original_data) %in% columns_to_remove)]
```

```
names(data) <- c("Age", "Day", "Month", "Region", "Temperature",
  "AtmPressure", "Precipitations",
  "Humidity", "Unemploymentrate", "Inflationrate")
```

```
names(data)
```

```
## [1] "Age" "Day" "Month"
## [4] "Region" "Temperature" "AtmPressure"
## [7] "Precipitations" "Humidity" "Unemploymentrate"
## [10] "Inflationrate"
```

```
#check for duplicates
```

```
duplicates <- duplicated(original_data)
```

```
# Print the duplicated rows
```

```
print("Duplicated rows:")
```

```
## [1] "Duplicated rows:"
```

```
print(original_data[duplicates, ])
```

```
## # A tibble: 0 x 21
## # i 21 variables: Seks <chr>, Age <chr>, Day_of_death <chr>,
## # Month_of_death <chr>, Year_of_death <chr>, Municipality_code <chr>,
## # Region <chr>, Province <chr>, Municipality <chr>,
## # Diagnostic_1_(cause_of_death) <chr>, Diagnostic_2_code <chr>,
## # Diagnostic_2_(suicide_method) <chr>, Place_of_occurrence <chr>,
## # Lowest_Temperature <chr>, Highest_Temperature <chr>,
## # Mean_temperature <chr>, Athmosferic_pressure_(mean)_in_hPa <chr>, ...
```

The variables of the data frame have correct values, without any inconsistencies regarding logical range of values. No duplicated rows found either.

We care about the following columns: Age and

Regarding

1. Economic factors: Unemployment rate (%) and Inflation rate(%)
2. Meteorological conditions: Mean Temperature, Atmospheric pressure (mean) in hPa, Precipitations(mm), Humidity (%).

```
#function that takes a column (col) as input
#If col is numeric, it rounds all elements of the vector to one decimal
#If col is not numeric, it converts its elements to numeric u
#If any element cannot be converted to numeric, a warning message
#is printed indicating that there are NA values function with one argument.
convert_and_round <- function(col) {
  if (is.numeric(col)) {
    round(col, 1)
  } else {
    numeric_col <- as.numeric(as.character(col))
    if (any(is.na(numeric_col))) {
      message("Warning: NA values in column ", names(col))
    }
    round(numeric_col, 1)
  }
}

data[, c(1, 5:8)] <- sapply(data[, c(1, 5:8)], convert_and_round)
```

```
## Warning: NA values in column
## Warning: NA values in column
## Warning: NA values in column
## Warning: NA values in column
```

```
data$Unemploymentrate <- round(as.numeric(data$Unemploymentrate),3)
data$Inflationrate <- round(as.numeric(data$Inflationrate),3)
```

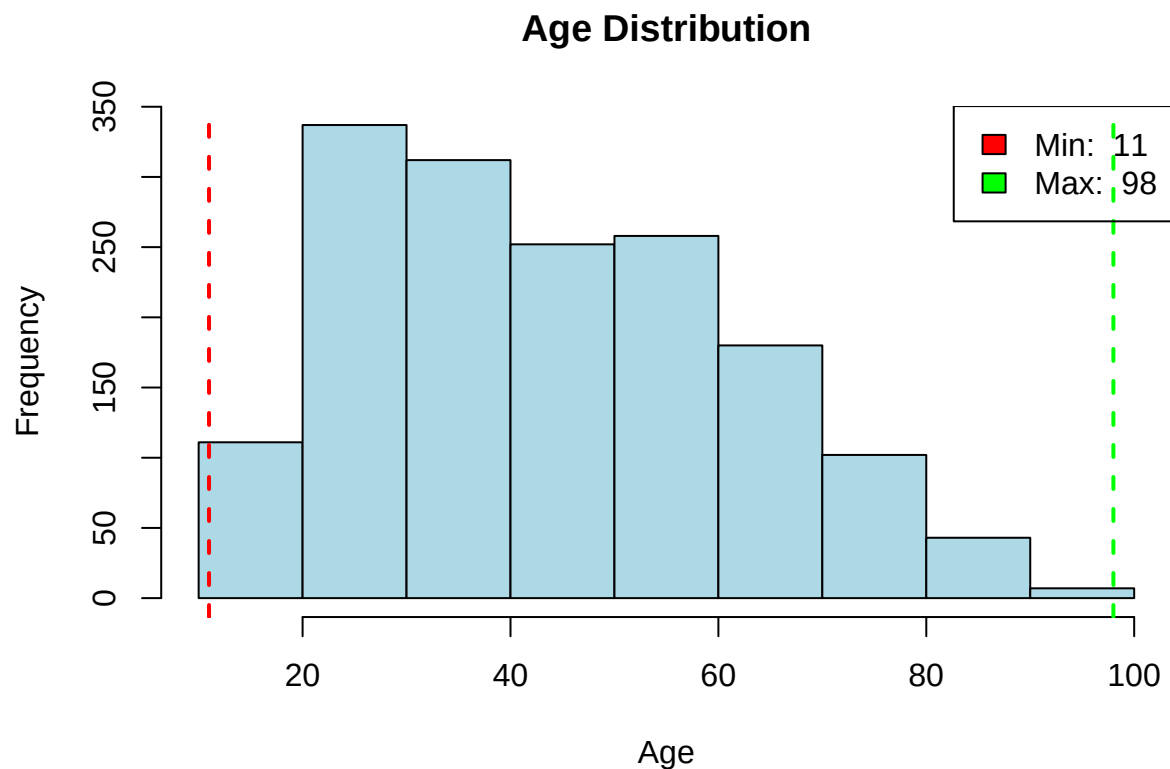
```
names(data)
```

```
## [1] "Age"           "Day"           "Month"
## [4] "Region"        "Temperature"   "AtmPressure"
## [7] "Precipitations" "Humidity"      "Unemploymentrate"
## [10] "Inflationrate"
```

Explore Data

```
# Calculate minimum and maximum values
min_age <- min(data$Age)
max_age <- max(data$Age)
```

```
# Plotting the age distribution with min and max values
hist(data$Age,
     main="Age Distribution",
     xlab="Age",
     ylab="Frequency",
     col="lightblue",
     border="black")
abline(v = min_age, col = "red", lwd = 2, lty = 2)
abline(v = max_age, col = "green", lwd = 2, lty = 2)
legend("topright", legend=c(paste("Min: ", min_age),
                             paste("Max: ", max_age)),
      fill=c("red", "green"))
```



```
# Adding the missing value in unemployment rate given by client
which(is.na(data$Unemploymentrate))

## [1] 1351

#data[1352,] # 0.108 = 10.76%

# Assuming data is your data frame
data[is.na(data$Unemploymentrate), "Unemploymentrate"] <- 0.108

which(is.na(data$Unemploymentrate))
```

```
## integer(0)
```

```
head(data)
```

```
## # A tibble: 6 x 10
##   Age Day  Month Region      Temperature AtmPressure Precipitations Humidity
##   <dbl> <chr> <chr> <chr>          <dbl>         <dbl>         <dbl>      <dbl>
## 1    20 1     1     Los Lagos      15.7         1004.          0         70
## 2    62 1     1     Metropolitana~ 22.3          954.          0         41
## 3    63 1     1     Los Lagos      16.4         1006.          0        64.5
## 4    57 1     1     Biobío         15.1         1013.          0         82
## 5    53 1     1     La Araucanía    18           1001.          0        59.8
## 6    51 1     1     Biobío         15.1         1013.          0         82
## # i 2 more variables: Unemploymentrate <dbl>, Inflationrate <dbl>
```

```
#Add new column to dataset with name of age group
```

```
data$AgeGroup <- cut(
  data$Age,
  breaks = c(0, 26, 40, 60, Inf),
  labels = c("Adolescents", "Young_adults", "Middle_adults", "Elderly_adults"),
  include.lowest = TRUE
)
```

```
#check how many observations per age group
```

```
sum(data$AgeGroup== "Elderly_adults")
```

```
## [1] 332
```

```
sum(data$AgeGroup== "Adolescents")
```

```
## [1] 305
```

```
sum(data$AgeGroup=="Young_adults")
```

```
## [1] 455
```

```
sum(data$AgeGroup=="Middle_adults")
```

```
## [1] 510
```

Boxplots to check distributions and outliers of variables

```
names(data)
```

```
## [1] "Age"           "Day"           "Month"
## [4] "Region"        "Temperature"    "AtmPressure"
## [7] "Precipitations" "Humidity"       "Unemploymentrate"
## [10] "Inflationrate" "AgeGroup"
```

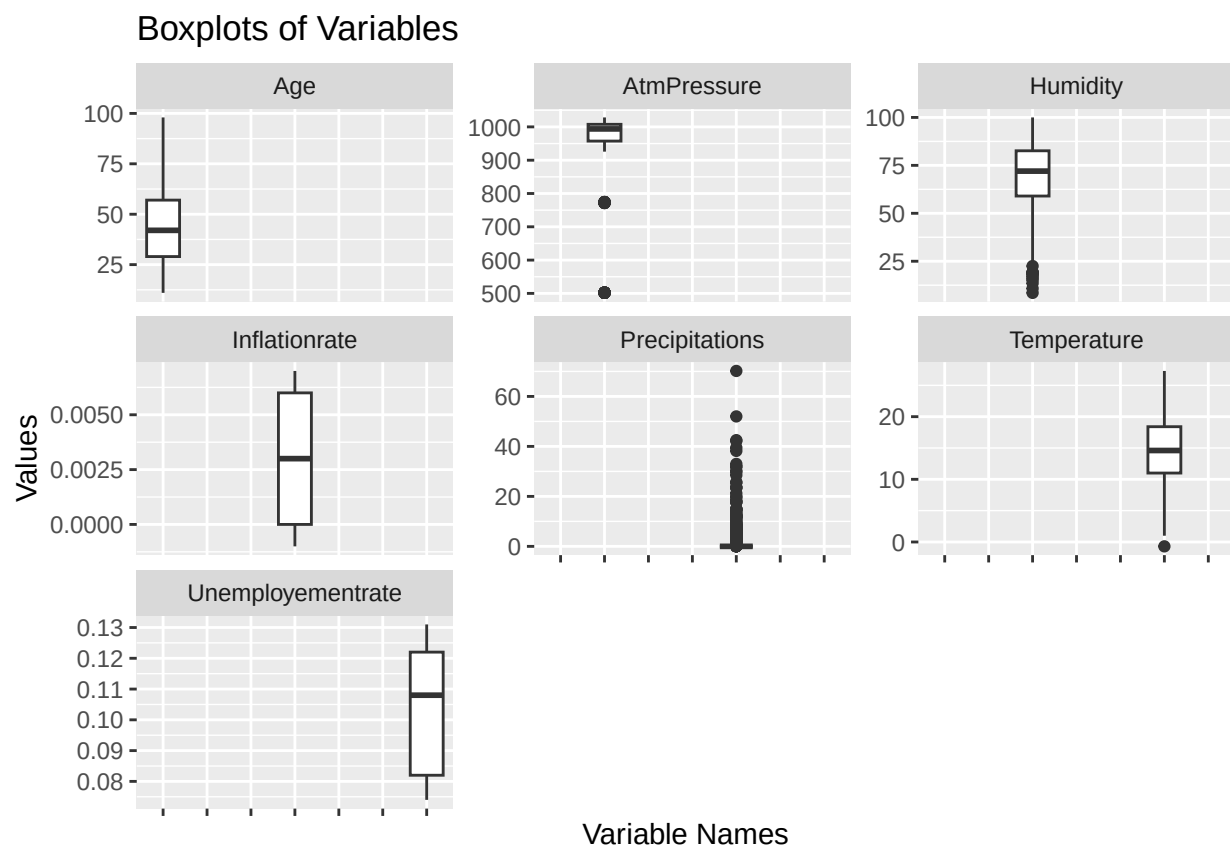
```
data_plot=data[, !names(data) %in% c("Day", "Month", "Region",
                                     "AgeGroup")]
```

```
data_long <- gather(data_plot, key = "Variable", value = "Value")
```

```
gg=ggplot(data_long, aes(x = Variable, y = Value)) +
  geom_boxplot() +
  labs(x = "Variable Names", y = "Values") +
  ggtitle("Boxplots of Variables") +
  facet_wrap(~ Variable, scales = "free_y", nrow=3, ncol =3)
```

```
gg + theme(axis.text.x = element_blank())
```

```
## Warning: Removed 458 rows containing non-finite values (‘stat_boxplot()’).
```



Outliers

```
count_outliers <- function(data) {
  outliers_count <- data.frame(Variable = character(), Outliers = numeric())

  for (col in names(data)) {
    if (is.numeric(data[[col]])) {
```

```

# Exclude NA values when calculating quartiles and identifying outliers
numeric_values <- na.omit(data[[col]])
Q1 <- quantile(numeric_values, 0.25)
Q3 <- quantile(numeric_values, 0.75)
IQR <- Q3 - Q1

lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

outliers <- numeric_values[numeric_values < lower_bound
                           | numeric_values > upper_bound]
num_outliers <- length(outliers)

outliers_count <- rbind(outliers_count, data.frame(Variable = col,
                                                    Outliers = num_outliers))
}
}

return(outliers_count)
}

# Usage example:
# Assuming your dataframe is named "data"
outliers_summary <- count_outliers(data)
print(outliers_summary)

```

```

##           Variable Outliers
## 1           Age           0
## 2    Temperature           1
## 3    AtmPressure          46
## 4  Precipitations        236
## 5      Humidity          12
## 6 Unemploymentrate           0
## 7    Inflationrate           0

```

-Atmospheric Presure: Some days with very low Atmospheric pressure (unsettled weather conditions).

-Humidity: Some days with very low Humidity

-Mean temperature: One outlier: -0.7

-Precipitations: Many outliers indicating that most of the values in the precipitation column are zeros , meaning days with no rain,snow etc.

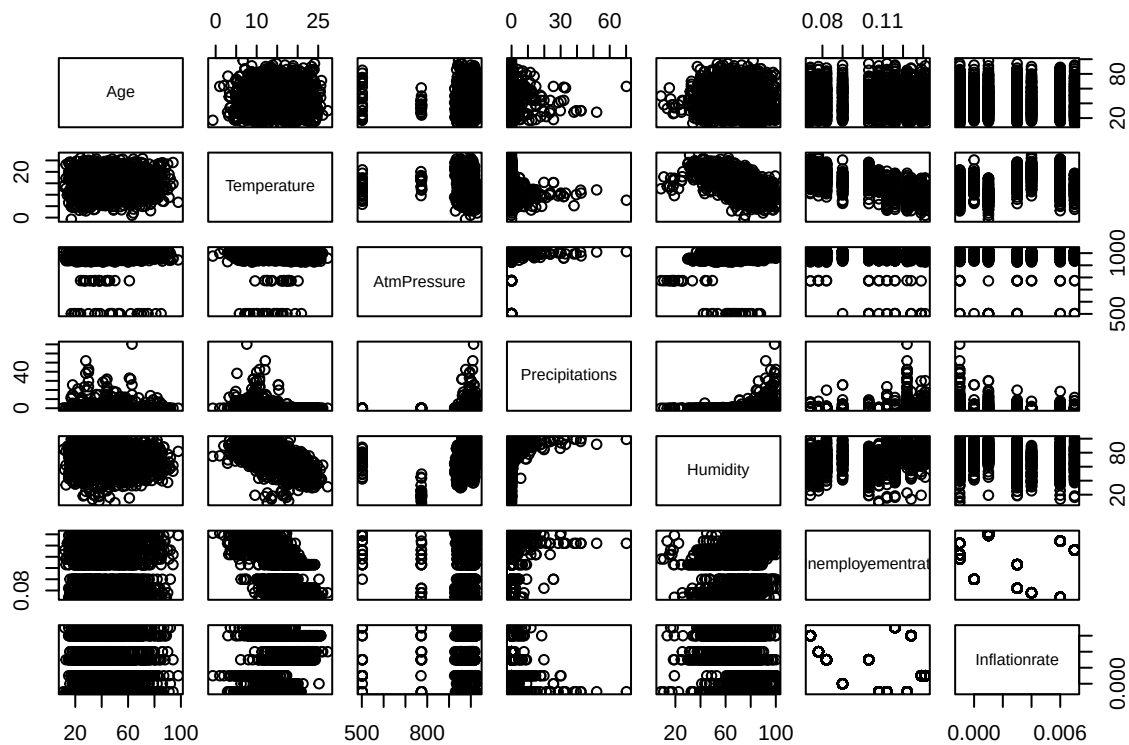
Skewness: Atmospheric Presure and Unemployment rate are is negative skewed. The other variables are not skewed.

Correlations:

```

pairs(data_plot)

```



Missing values

```
#Whole dataset
miss_var_summary(data)
```

```
## # A tibble: 11 x 3
##   variable      n_miss pct_miss
##   <chr>         <int>   <dbl>
## 1 Precipitations    368    23.0
## 2 AtmPressure       32     2.00
## 3 Temperature       29     1.81
## 4 Humidity          29     1.81
## 5 Age               0      0
## 6 Day               0      0
## 7 Month             0      0
## 8 Region            0      0
## 9 Unemploymentrate  0      0
## 10 Inflationrate    0      0
## 11 AgeGroup         0      0
```

```
names(data)
```

```
## [1] "Age"           "Day"           "Month"
```

```
## [4] "Region"          "Temperature"      "AtmPressure"
## [7] "Precipitations"   "Humidity"         "Unemploymentrate"
## [10] "Inflationrate"    "AgeGroup"
```

Multiple Imputation

```
names(data)
```

```
## [1] "Age"          "Day"          "Month"
## [4] "Region"       "Temperature"   "AtmPressure"
## [7] "Precipitations" "Humidity"     "Unemploymentrate"
## [10] "Inflationrate" "AgeGroup"
```

```
# Assuming you have already performed multiple imputations
```

```
data$Region=as.factor(data$Region)
```

```
imp <- mice(
  data = data,
  method = 'rf',
  pred = quickpred(data, mincor = 0.1,
    exclude = c("Age",
      "Unemploymentrate",
      "Inflationrate",
      "AgeGroup",
      "Day",
      "Month")))
```

```
##
## iter imp variable
## 1 1 Temperature AtmPressure Precipitations Humidity
## 1 2 Temperature AtmPressure Precipitations Humidity
## 1 3 Temperature AtmPressure Precipitations Humidity
## 1 4 Temperature AtmPressure Precipitations Humidity
## 1 5 Temperature AtmPressure Precipitations Humidity
## 2 1 Temperature AtmPressure Precipitations Humidity
## 2 2 Temperature AtmPressure Precipitations Humidity
## 2 3 Temperature AtmPressure Precipitations Humidity
## 2 4 Temperature AtmPressure Precipitations Humidity
## 2 5 Temperature AtmPressure Precipitations Humidity
## 3 1 Temperature AtmPressure Precipitations Humidity
## 3 2 Temperature AtmPressure Precipitations Humidity
## 3 3 Temperature AtmPressure Precipitations Humidity
## 3 4 Temperature AtmPressure Precipitations Humidity
## 3 5 Temperature AtmPressure Precipitations Humidity
## 4 1 Temperature AtmPressure Precipitations Humidity
## 4 2 Temperature AtmPressure Precipitations Humidity
## 4 3 Temperature AtmPressure Precipitations Humidity
## 4 4 Temperature AtmPressure Precipitations Humidity
## 4 5 Temperature AtmPressure Precipitations Humidity
## 5 1 Temperature AtmPressure Precipitations Humidity
## 5 2 Temperature AtmPressure Precipitations Humidity
```



```
## 5 3 Temperature AtmPressure Precipitations Humidity
## 5 4 Temperature AtmPressure Precipitations Humidity
## 5 5 Temperature AtmPressure Precipitations Humidity
```

Functions to transform dataset in the appropriate way to perform poisson regression

```
# Function to z-score the data
z_scores_fun <- function(imputed_data) {
  data_zscored <- imputed_data %>%
    group_by(Region) %>%
    mutate(
      zs_temp = scale(Temperature),
      zs_atm_press = scale(AtmPressure),
      zs_Preci = scale(Precipitations),
      zs_humi = scale(Humidity)
    )
  return(data_zscored)
}

# Function to process suicide data by day
process_suicide_data_by_day_r <- function(data) {
  # Convert Month_of_death and Day_of_death to Date format
  data <- data %>%
    mutate(Date = as.Date(paste("2020", Month, Day, sep="-"), format="%Y-%m-%d"))

  # Group data by Date and Region, then calculate summary statistics
  suicides_by_day_r <- data %>%
    group_by(Date, Region, AgeGroup) %>%
    summarise(
      Number_of_suicides = n(),
      z_temperature = mean(zs_temp),
      z_Atmospheric_pressure = mean(zs_atm_press),
      z_Precipitations_mm = mean(zs_Preci),
      z_Humidity_percentage = mean(zs_humi),
      .groups = "drop"
    )

  return(suicides_by_day_r)
}

# Apply z-score and process data for each imputed dataset
# Five complete datasets are created with the z_scores of the metereological factors
processed_datasets <- lapply(1:5, function(i) {
  imp_day <- complete(imp, action = i)
  zscored_data <- z_scores_fun(imp_day)
  processed_data <- process_suicide_data_by_day_r(zscored_data)
  return(processed_data)
})
```

Poisson Rgression

Metereological Conditions - models with Regions as Random effects

Models are fitted with all the imputed datasets and the results are pooled

Fiting the model without taking into account the age

```
# The z scores of precipitation equal to zero were calculated
# as NaN sto this is corrected here
data_z_scored=z_scores_fun(complete(imp))
data_z_scored$zs_Preci[is.na(data_z_scored$zs_Preci)] <- 0
```

```
model_meter_no_age=glmer(Number_of_suicides ~ z_temperature +
                          z_Atmospheric_pressure +
                          z_Precipitations_mm +
                          z_Humidity_percentage +
                          (1|Region),
                          family = poisson,
                          data = process_suicide_data_by_day_r(data_z_scored))
summary(model_meter_no_age)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: poisson ( log )
## Formula: Number_of_suicides ~ z_temperature + z_Atmospheric_pressure +
##          z_Precipitations_mm + z_Humidity_percentage + (1 | Region)
## Data: process_suicide_data_by_day_r(data_z_scored)
##
##      AIC      BIC   logLik deviance df.resid
## 3144.1   3176.0 -1566.1  3132.1     1479
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -0.13696 -0.08734 -0.07160 -0.05867  2.76953
##
## Random effects:
## Groups Name          Variance Std.Dev.
## Region (Intercept) 0.0004122 0.0203
## Number of obs: 1485, groups: Region, 16
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    7.251e-02  2.936e-02   2.469   0.0135 *
## z_temperature    1.844e-02  3.196e-02   0.577   0.5639
## z_Atmospheric_pressure 4.863e-03  2.583e-02   0.188   0.8507
## z_Precipitations_mm    5.526e-03  2.671e-02   0.207   0.8361
## z_Humidity_percentage 6.025e-05  3.289e-02   0.002   0.9985
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##          (Intr) z_tmpr z_Atm_ z_Prc_
## z_temperatr -0.006
## z_Atmsphrc_  0.001  0.034
## z_Prcpttns_ -0.001  0.025  0.121
## z_Hmdty_prc  0.001  0.587 -0.148 -0.214
```

```
ranef(model_meter_no_age)
```

```
## $Region
##          (Intercept)
## Antofagasta          -0.0010195601
## Arica y Parinacota    -0.0001340025
## Atacama               -0.0007982405
## Aysén del General Carlos Ibáñez del Campo -0.0003402634
## Biobío                -0.0004893531
## Coquimbo              -0.0017980231
## La Araucanía          -0.0021108714
## Libertador General Bernardo O'Higgins    -0.0016044781
## Los Lagos             -0.0014759460
## Los Ríos              -0.0015783922
## Magallanes y de la Antártica Chilena      -0.0007684078
## Maule                 -0.0010014831
## Metropolitana de Santiago  0.0152272342
## Ñuble                 -0.0012936612
## Tarapacá              -0.0008889346
## Valparaíso            0.0001916205
##
## with conditional variances for "Region"
```

Fiting the model taking into account the age

```
model_meter=glmer(Number_of_suicides ~ AgeGroup:(z_temperature
+ z_Atmospheric_pressure
+ z_Precipitations_mm
+ z_Humidity_percentage)
+ (1|Region),
family = poisson,
data = process_suicide_data_by_day_r(data_z_scored))
summary(model_meter)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: poisson ( log )
## Formula:
## Number_of_suicides ~ AgeGroup:(z_temperature + z_Atmospheric_pressure +
## z_Precipitations_mm + z_Humidity_percentage) + (1 | Region)
## Data: process_suicide_data_by_day_r(data_z_scored)
##
##      AIC      BIC    logLik deviance df.resid
##  3167.6   3263.1  -1565.8   3131.6     1467
##      27
```

```

##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -0.21367 -0.08866 -0.07220 -0.05432  2.75007
##
## Random effects:
##   Groups Name      Variance Std.Dev.
##   Region (Intercept) 0.0004039 0.0201
## Number of obs: 1485, groups: Region, 16
##
## Fixed effects:
##                                     Estimate Std. Error z value
## (Intercept)                      0.0730418  0.0294760   2.478
## AgeGroupAdolescents:z_temperature 0.0325097  0.0717509   0.453
## AgeGroupYoung_adults:z_temperature 0.0294708  0.0573171   0.514
## AgeGroupMiddle_adults:z_temperature 0.0111844  0.0617907   0.181
## AgeGroupElderly_adults:z_temperature 0.0009665  0.0686928   0.014
## AgeGroupAdolescents:z_Atmospheric_pressure 0.0157400  0.0657345   0.239
## AgeGroupYoung_adults:z_Atmospheric_pressure -0.0121954  0.0506272  -0.241
## AgeGroupMiddle_adults:z_Atmospheric_pressure 0.0161431  0.0420577   0.384
## AgeGroupElderly_adults:z_Atmospheric_pressure -0.0046282  0.0601101  -0.077
## AgeGroupAdolescents:z_Precipitations_mm 0.0133380  0.0876724   0.152
## AgeGroupYoung_adults:z_Precipitations_mm 0.0008294  0.0425856   0.019
## AgeGroupMiddle_adults:z_Precipitations_mm -0.0013873  0.0475365  -0.029
## AgeGroupElderly_adults:z_Precipitations_mm 0.0192162  0.0633822   0.303
## AgeGroupAdolescents:z_Humidity_percentage -0.0068271  0.0737583  -0.093
## AgeGroupYoung_adults:z_Humidity_percentage 0.0134274  0.0586083   0.229
## AgeGroupMiddle_adults:z_Humidity_percentage -0.0023191  0.0642413  -0.036
## AgeGroupElderly_adults:z_Humidity_percentage -0.0102344  0.0710224  -0.144
##                                     Pr(>|z|)
## (Intercept)                      0.0132 *
## AgeGroupAdolescents:z_temperature 0.6505
## AgeGroupYoung_adults:z_temperature 0.6071
## AgeGroupMiddle_adults:z_temperature 0.8564
## AgeGroupElderly_adults:z_temperature 0.9888
## AgeGroupAdolescents:z_Atmospheric_pressure 0.8108
## AgeGroupYoung_adults:z_Atmospheric_pressure 0.8096
## AgeGroupMiddle_adults:z_Atmospheric_pressure 0.7011
## AgeGroupElderly_adults:z_Atmospheric_pressure 0.9386
## AgeGroupAdolescents:z_Precipitations_mm 0.8791
## AgeGroupYoung_adults:z_Precipitations_mm 0.9845
## AgeGroupMiddle_adults:z_Precipitations_mm 0.9767
## AgeGroupElderly_adults:z_Precipitations_mm 0.7618
## AgeGroupAdolescents:z_Humidity_percentage 0.9263
## AgeGroupYoung_adults:z_Humidity_percentage 0.8188
## AgeGroupMiddle_adults:z_Humidity_percentage 0.9712
## AgeGroupElderly_adults:z_Humidity_percentage 0.8854
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it

```

```
random_effects1 <- ranef(model_meter)
print(random_effects1)
```

```
## $Region
## (Intercept)
## Antofagasta -1.004550e-03
## Arica y Parinacota -8.919670e-05
## Atacama -7.694009e-04
## Aysén del General Carlos Ibáñez del Campo -3.290579e-04
## Biobío -5.118920e-04
## Coquimbo -1.918085e-03
## La Araucanía -2.050003e-03
## Libertador General Bernardo O'Higgins -1.581084e-03
## Los Lagos -1.391720e-03
## Los Ríos -1.564981e-03
## Magallanes y de la Antártica Chilena -7.387274e-04
## Maule -9.412407e-04
## Metropolitana de Santiago 1.493220e-02
## Ñuble -1.183792e-03
## Tarapacá -8.282292e-04
## Valparaíso 8.561782e-05
##
## with conditional variances for "Region"
```

Economic factors

Creation of dataframe with number of suicides for all the Months

```
process_suicide_data_by_month <- function(data) {
  # Convert Month_of_death to Date format
  data <- data %>%
    mutate(Date = as.Date(paste("2020", Month, "01", sep="-"), format="%Y-%m-%d"))

  # Group data by Month and calculate summary statistics
  suicides_by_month <- data %>%
    group_by(Year = lubridate::year(Date), Month = lubridate::month(Date), AgeGroup) %>%
    summarise(
      Number_of_suicides = n(),
      Inflation_rate_percentage = mean(Inflationrate),
      Unemployment_rate_percentage = mean(Unemploymentrate),
      .groups = 'drop' # Prevent automatic grouping by 'Year'
    )

  return(suicides_by_month)
}

suicides_by_month=process_suicide_data_by_month(data)
names(suicides_by_month)
```

```
## [1] "Year" "Month"
## [3] "AgeGroup" "Number_of_suicides"
## [5] "Inflation_rate_percentage" "Unemployment_rate_percentage"
```

Fiting the model without taking into account the age

```
glm_model_econ_no_age <- glm(Number_of_suicides ~ Inflation_rate_percentage +
                             Unemployment_rate_percentage ,
                             family = poisson,
                             data = suicides_by_month)

summary(glm_model_econ_no_age)

##
## Call:
## glm(formula = Number_of_suicides ~ Inflation_rate_percentage +
##      Unemployment_rate_percentage, family = poisson, data = suicides_by_month)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.5187  -1.3549  -0.2493   0.9480   3.3700
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.8970     0.1459  26.711 < 2e-16 ***
## Inflation_rate_percentage  31.7419     9.2294   3.439 0.000583 ***
## Unemployment_rate_percentage -4.4748     1.3052  -3.428 0.000607 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 140.61  on 47  degrees of freedom
## Residual deviance: 108.45  on 45  degrees of freedom
## AIC: 369.24
##
## Number of Fisher Scoring iterations: 4
```

Fiting the model taking into account the age

This model includes a joint interaction term for “AgeGroup” with both “Inflation_rate_percentage” and “Unemployment_rate_percentage”. The inclusion of these interaction terms allows the models to capture potential non-additive effects or interactions between the predictors.

```
suicides_by_month$AgeGroup=as.factor(suicides_by_month$AgeGroup)
glm_model_econ <- glm(Number_of_suicides ~ Inflation_rate_percentage +
                     Unemployment_rate_percentage +
                     AgeGroup:(Inflation_rate_percentage +
                                Unemployment_rate_percentage),
                     family = poisson,
                     data = suicides_by_month)

summary(glm_model_econ)
```

```
##
```

```
## Call:
## glm(formula = Number_of_suicides ~ Inflation_rate_percentage +
##      Unemployment_rate_percentage + AgeGroup:(Inflation_rate_percentage +
##      Unemployment_rate_percentage), family = poisson, data = suicides_by_month)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.0080  -0.6527  -0.1055   0.5092   1.9984
##
## Coefficients:
##                                     Estimate Std. Error
## (Intercept)                        3.9188     0.1460
## Inflation_rate_percentage          13.2521     19.6602
## Unemployment_rate_percentage       -6.7665     1.4525
## Inflation_rate_percentage:AgeGroupYoung_adults  14.2100     24.8707
## Inflation_rate_percentage:AgeGroupMiddle_adults 33.8502     24.3143
## Inflation_rate_percentage:AgeGroupElderly_adults 17.3653     26.5422
## Unemployment_rate_percentage:AgeGroupYoung_adults  3.3822     0.8976
## Unemployment_rate_percentage:AgeGroupMiddle_adults  3.8968     0.8872
## Unemployment_rate_percentage:AgeGroupElderly_adults 0.4332     0.9679
##                                     z value Pr(>|z|)
## (Intercept)                        26.843 < 2e-16 ***
## Inflation_rate_percentage           0.674 0.500276
## Unemployment_rate_percentage       -4.659 3.18e-06 ***
## Inflation_rate_percentage:AgeGroupYoung_adults  0.571 0.567760
## Inflation_rate_percentage:AgeGroupMiddle_adults  1.392 0.163864
## Inflation_rate_percentage:AgeGroupElderly_adults 0.654 0.512948
## Unemployment_rate_percentage:AgeGroupYoung_adults  3.768 0.000165 ***
## Unemployment_rate_percentage:AgeGroupMiddle_adults  4.392 1.12e-05 ***
## Unemployment_rate_percentage:AgeGroupElderly_adults 0.448 0.654453
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 140.606  on 47  degrees of freedom
## Residual deviance:  40.231  on 39  degrees of freedom
## AIC: 313.01
##
## Number of Fisher Scoring iterations: 4
```

Diagnostics

Metereological Fctors

Model without age

```
# Degrees of freedom
df <- df.residual(model_meter_no_age)

# Goodness-of-fit tests
```

```

observed_table <- process_suicide_data_by_day_r(data_z_scored)$Number_of_suicides
expected_table <- predict(model_meter_no_age, type = "response")
if (length(expected_table) != length(observed_table)) {
  stop("Length mismatch between observed and expected values.")
}
## Deviance
chi_square_deviance=deviance(model_meter_no_age)
# P-value
p_value_deviance <- pchisq(chi_square_deviance, df, lower.tail=F)

## Pearson
# Pearson residuals for overdispersion test
pearson_residuals <- residuals(model_meter_no_age, type = "pearson")
# Pearson statistic for overdispersion test
pearson_statistic <- sum(pearson_residuals^2)
p_value_pearson = pchisq(pearson_statistic, df, lower.tail=F)
# Degrees of freedom for overdispersion test

# Print results
output <- paste(
  "Goodness-of-Fit Tests\n",
  "Test\t\t DF\t Estimate\t Mean\t\t Chi-Square\t P-Value\n",
  "Deviance\t", df, "\t",
  round(chi_square_deviance, 4), "\t",
  round(chi_square_deviance/df, 4), "\t",
  round(chi_square_deviance, 4), "\t",
  round(p_value_deviance, 4), "\n",
  "Pearson\t\t", df, "\t",
  round(pearson_statistic, 4), "\t",
  round(pearson_statistic/df, 4), "\t",
  round(pearson_statistic, 4), "\t",
  round(p_value_pearson, 4), "\n"
)

# Check for lack-of-fit
if (p_value_deviance > 0.05) {
  output <- paste(output, "No evidence of lack-of-fit.\n")
} else {
  output <- paste(output, "Statistically significant lack of fit.\n")
}

if (p_value_pearson > 0.05) {
  output <- paste(output, "No evidence of overdispersion.\n")
} else {
  output <- paste(output, "Statistically significant overdispersion.\n")
}

cat(output)

```

```

## Goodness-of-Fit Tests
## Test      DF    Estimate    Mean      Chi-Square  P-Value
## Deviance  1479    91.7754    0.0621    91.7754    1
## Pearson   1479    119.1606   0.0806    119.1606    1

```



```
## No evidence of lack-of-fit.
## No evidence of overdispersion.
```

Model with age

```
# Degrees of freedom
df <- df.residual(model_meter)

# Goodness-of-fit tests
observed_table <- process_suicide_data_by_day_r(data_z_scored)$Number_of_suicides
expected_table <- predict(model_meter, type = "response")
if (length(expected_table) != length(observed_table)) {
  stop("Length mismatch between observed and expected values.")
}

## Deviance
chi_square_deviance=deviance(model_meter)
# P-value
p_value_deviance <- pchisq(chi_square_deviance, df, lower.tail=F)

## Pearson
# Pearson residuals for overdispersion test
pearson_residuals <- residuals(model_meter, type = "pearson")
# Pearson statistic for overdispersion test
pearson_statistic <- sum(pearson_residuals^2)
p_value_pearson = pchisq(pearson_statistic, df, lower.tail=F)
# Degrees of freedom for overdispersion test

# Print results
output <- paste(
  "Goodness-of-Fit Tests\n",
  "Test\t\t DF\t Estimate\t Mean\t\t Chi-Square\t P-Value\n",
  "Deviance\t", df, "\t",
  round(chi_square_deviance, 4), "\t",
  round(chi_square_deviance/df, 4), "\t",
  round(chi_square_deviance, 4), "\t",
  round(p_value_deviance, 4), "\n",
  "Pearson\t\t", df, "\t",
  round(pearson_statistic, 4), "\t",
  round(pearson_statistic/df, 4), "\t",
  round(pearson_statistic, 4), "\t",
  round(p_value_pearson, 4), "\n"
)

# Check for lack-of-fit
if (p_value_deviance > 0.05) {
  output <- paste(output, "No evidence of lack-of-fit.\n")
} else {
  output <- paste(output, "Statistically significant lack of fit.\n")
}

if (p_value_pearson > 0.05) {
  output <- paste(output, "No evidence of overdispersion.\n")
}
```

```

} else {
  output <- paste(output, "Statistically significant overdispersion.\n")
}

cat(output)

```

```

## Goodness-of-Fit Tests
## Test      DF    Estimate    Mean      Chi-Square  P-Value
## Deviance   1467    91.275     0.0622     91.275      1
## Pearson    1467   118.2671    0.0806    118.2671     1
## No evidence of lack-of-fit.
## No evidence of overdispersion.

```

Economic Indicators

Model without age

```

# Degrees of freedom
df <- df.residual(glm_model_econ_no_age)

# Goodness-of-fit tests
observed_table <- suicides_by_month$Number_of_suicides
expected_table <- predict(glm_model_econ_no_age, type = "response")
if (length(expected_table) != length(observed_table)) {
  stop("Length mismatch between observed and expected values.")
}

## Deviance
chi_square_deviance=deviance(glm_model_econ_no_age)
# P-value
p_value_deviance <- pchisq(chi_square_deviance, df, lower.tail=F)

## Pearson
# Pearson residuals for overdispersion test
pearson_residuals <- residuals(glm_model_econ_no_age, type = "pearson")
# Pearson statistic for overdispersion test
pearson_statistic <- sum(pearson_residuals^2)
p_value_pearson = pchisq(pearson_statistic, df, lower.tail=F)
# Degrees of freedom for overdispersion test

# Print results
output <- paste(
  "Goodness-of-Fit Tests\n",
  "Test\t\t DF\t Estimate\t Mean\t\t Chi-Square\t P-Value\n",
  "Deviance\t", df, "\t",
  round(chi_square_deviance, 4), "\t",
  round(chi_square_deviance/df, 4), "\t",
  round(chi_square_deviance, 4), "\t",
  round(p_value_deviance, 4), "\n",
  "Pearson\t\t", df, "\t",
  round(pearson_statistic, 4), "\t",
  round(pearson_statistic/df, 4), "\t",

```

```

    round(pearson_statistic, 4), "\t",
    round(p_value_pearson, 4), "\n"
)

# Check for lack-of-fit
if (p_value_deviance > 0.05) {
  output <- paste(output, "No evidence of lack-of-fit.\n")
} else {
  output <- paste(output, "Statistically significant lack of fit.\n")
}

if (p_value_pearson > 0.05) {
  output <- paste(output, "No evidence of overdispersion.\n")
} else {
  output <- paste(output, "Statistically significant overdispersion.\n")
}

cat(output)

```

```

## Goodness-of-Fit Tests
## Test      DF   Estimate    Mean      Chi-Square  P-Value
## Deviance   45    108.4543    2.4101    108.4543    0
## Pearson   45    111.3796    2.4751    111.3796    0
## Statistically significant lack of fit.
## Statistically significant overdispersion.

```

Model with age

```

# Degrees of freedom
df <- df.residual(glm_model_econ)

# Goodness-of-fit tests
observed_table <- suicides_by_month$Number_of_suicides
expected_table <- predict(glm_model_econ, type = "response")
if (length(expected_table) != length(observed_table)) {
  stop("Length mismatch between observed and expected values.")
}

## Deviance
chi_square_deviance=deviance(glm_model_econ)
# P-value
p_value_deviance <- pchisq(chi_square_deviance, df, lower.tail=F)

## Pearson
# Pearson residuals for overdispersion test
pearson_residuais <- residuals(glm_model_econ, type = "pearson")
# Pearson statistic for overdispersion test
pearson_statistic <- sum(pearson_residuais^2)
p_value_pearson = pchisq(pearson_statistic, df, lower.tail=F)
# Degrees of freedom for overdispersion test

# Print results

```

```

output <- paste(
  "Goodness-of-Fit Tests\n",
  "Test\t\t DF\t Estimate\t Mean\t\t Chi-Square\t P-Value\n",
  "Deviance\t", df, "\t",
  round(chi_square_deviance, 4), "\t",
  round(chi_square_deviance/df, 4), "\t",
  round(chi_square_deviance, 4), "\t",
  round(p_value_deviance, 4), "\n",
  "Pearson\t\t", df, "\t",
  round(pearson_statistic, 4), "\t",
  round(pearson_statistic/df, 4), "\t",
  round(pearson_statistic, 4), "\t",
  round(p_value_pearson, 4), "\n"
)

# Check for lack-of-fit
if (p_value_deviance > 0.05) {
  output <- paste(output, "No evidence of lack-of-fit.\n")
} else {
  output <- paste(output, "Statistically significant lack of fit.\n")
}

if (p_value_pearson > 0.05) {
  output <- paste(output, "No evidence of overdispersion.\n")
} else {
  output <- paste(output, "Statistically significant overdispersion.\n")
}

cat(output)

```

```

## Goodness-of-Fit Tests
## Test      DF    Estimate    Mean      Chi-Square  P-Value
## Deviance   39     40.2311    1.0316     40.2311    0.4155
## Pearson   39     40.741     1.0446     40.741     0.3937
## No evidence of lack-of-fit.
## No evidence of overdispersion.

```

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